

CSS₃

What is CSS?

- CSS stands for Cascading Style Sheet
- Used to design or formatting HTML elements
- Widely used for webpages and web-applications
- CSS describes **How HTML Contents are to be displayed on webpages.**
- It controls layout of multiple webpages at once
- It maintains informality for same types of elements
- External stylesheet is stored as .css file

Ways to apply CSS

- External File

```
<head>
```

```
    <link rel="stylesheet" type="text/css" href="mystyle.css">
```

```
</head>
```

- Internal CSS

```
<style>
```

```
    CSS Code
```

```
</style>
```

- Inline CSS

```
<h1 style="color:blue;font-size:25px"> Inline CSS </h1>
```

CSS v/s CSS3

CSS	CSS3
CSS does not support media query for responsive webpage designing	CSS3 supports media query for responsive webpage designing
Can not split into varied modules	Can be easily split into varied modules
Does not support by all new browser	Support all new browsers
It has old and standard colours	It supports new RGBA, HSLA, HSL and gradient colours
Not mobile friendly	Mobile friendly
Can not support visual effects like shadowing, text selections, animation etc.	It supports visual effects and animations.
Can not has rounded border feature	It has rounded borders feature
2D and 3D transformation not supported	2D and 3D transformation not supported

Adding Borders

- CSS Border Properties

border-width

border-style

border-color

Example:

```
p{  
border-width:2px;  
border-style:solid;  
border-color red;  
}
```

Adding Borders

- Different Border Style

Solid

Dotted

Dashed

Double

Groove

Ridge

Inset

Outset

None

Adding Borders

- Border Shorthand

Border: width style color

Example:

```
p{  
border:2px solid red;  
}
```

Adding Borders

- Specific Borders for Individual Sides

`border-top`

`border-right`

`border-bottom`

`border-left`

Example:

```
p{  
border-bottom:2px solid red;  
}
```


Adding Borders

- You can directly add the border side into border style properties like

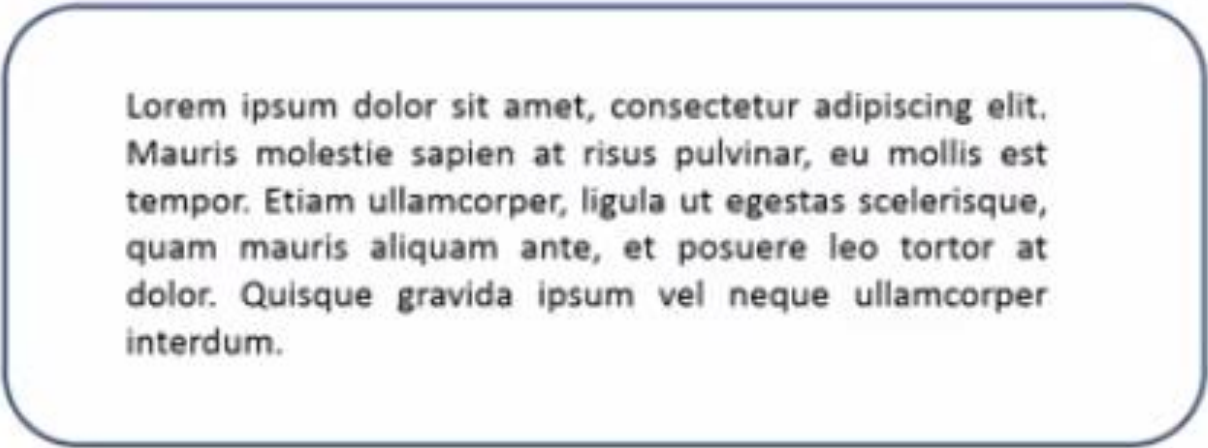
```
p{  
border-width:pixel;  
border-style:border-top border-right border-bottom border-left;  
border-color: color;  
}
```

Example:

```
p{  
border-width:2px;  
Border-style: dashed solid dotted double;  
}
```

border-radius

- Border-radius property is introduced in CC3
- It is used to give rounded corner to HTML object



Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris molestie sapien at risus pulvinar, eu mollis est tempor. Etiam ullamcorper, ligula ut egestas scelerisque, quam mauris aliquam ante, et posuere leo tortor at dolor. Quisque gravida ipsum vel neque ullamcorper interdum.

- In above example you can see div has rounded corner

border-radius

- border-radius
 - border-top-left-radius
 - border-top-right-radius
 - border-bottom-right-radius
 - border-bottom-left-radius

border-radius

■ border-radius Shorthand

Border-radius: border-top-left, border-top-right, border-bottom-right, border-bottom-left;

border-top-left

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border-top-right

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Mauris molestie sapien at risus pulvinar, eu mollis est tempor. Etiam ullamcorper, ligula ut egestas scelerisque, quam mauris aliquam ante, et posuere leo tortor at dolor. Quisque gravida ipsum vel neque ullamcorper interdum.

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border-bottom-right

border-bottom-left

border-radius: 20px, 20px, 20px, 20px;

Adding Background

- background-color
- background-image
- background-repeat
- background-position
- background-attachment
- background-size
- background-origin
- background-clip

background-color

background-color property is used to set the background color of HTML element

Syntax:

background-color: name of color;

or

background-color: #color-code;

background-image

background-image property is used to set the image as background in HTML element

Syntax:

background-image: Absolute or relative path;

or

background-image: url();

background-repeat

background-repeat property sets if/how a background image will be repeated. By default, a background-image is repeated both vertically and horizontally.

Syntax:

```
background-repeat: property-value;
```

Property Values:

```
background-repeat: repeat;
```

```
background-repeat: no-repeat;
```

```
background-repeat: repeat-x;
```

```
background-repeat: repeat-y;
```


background-repeat

Property Values (introduced in CSS3):

`background-repeat: space;`

The image is repeated as much as possible while avoiding clipping. The 1st and last copies of the image are placed to either side of the containing element.

`background-repeat: round;`

The images will stretch or shrink slightly to avoid clipping and to produce no gaps.

background-position

background-position property allows us to define where a background image should be positioned within an element.

This property can take length, percentage or keyword values.

Syntax:

background-position: x-axis_value y-axis_value;

The available keywords are **top**, **bottom**, **right**, **left** and **center**. The keywords can be used in any order, and when only one keyword is used the other is assumed to be set to center.

If an element has multiple background images, you can define a comma-separated list of background-position values in the same order as the background images are defined.

background-position

Syntax:

```
background-position: x% y%;  
background-position: Xpx Ypx;  
background-position: x-keyword y-keyword;
```

Examples:

```
background-position: 75% 0;  
background-position: 100%;  
background-position: 20px, 20px;  
background-position: 50%;  
background-position: 0 90%;
```

background-position

Syntax (For multiple images):

background-position: x% y%, Xpx Ypx, x-keyword y-keyword;

Examples:

background-position: 75% 0, 100%, 20px,20px, 50%,0 90%;

background-attachment

background-attachment property allows us to define where a background image should be scrolled with page content or it should be fixed on defined position (here content will be moved over image).

This property can take two values.

- Scrolled (default property, moved with content)
- Fixed (content will be moved over the image)

background-attachment

Syntax (For multiple images):

background-attachment: property-value;

Examples:

background-attachment: scrolled;

background-attachment: fixed;

background-size

background-size property introduced in CSS3 only. This property will allow us to set the size of background/image.

This property can take length, percentage or keyword values.

There are many variations and different syntaxes you can use for this property, same like background-position(except keyword).

There are two keyword you can use.

- cover
- contain

background-size

- When you use one value as a property value it will use value for width and set height as auto.
e.g. `background-size: 250px;` sets image width to 250px and keep height as auto.
- **cover:** tells the browser to make sure the image always covers the entire container, even if it has to stretch the image or cut a little bit off one of the edges.
- **Contain:** on the other hand, says to always show the whole image, even if that leaves a little space to the sides or bottom.

background-size

Syntax:

```
background-size: x% y%;  
background-size: Xpx Ypx;  
background-size: keyword;
```

Examples:

```
background-size: 350px 700px;  
background-size: 300px;  
background-size: 100% 50%;  
background-size: 50%;  
background-size: cover;  
background-size: contain;  
background-size: 350px 700px, contain;
```

background-origin

- **background-origin** is a new property introduced in CSS3
- The **background-origin** property specifies the origin position (the background positioning area) of a background image.

Syntax:

background-origin: value;

It has 3 values:

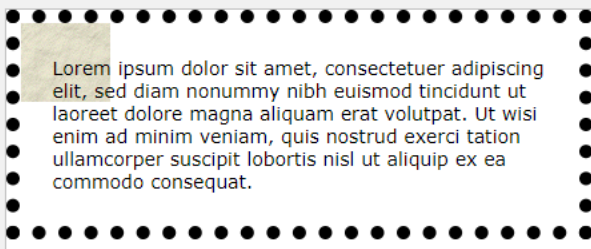
- **padding-box** :
- **border-box**
- **content-box**

background-origin

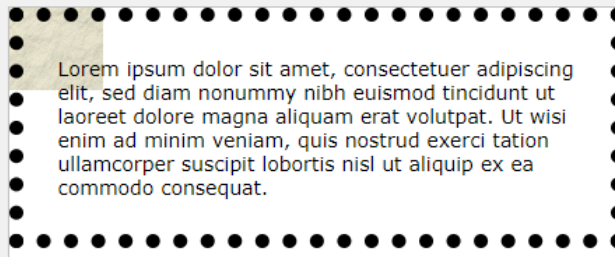
It has 3 values:

- **padding-box** : Default value. The background image starts from the upper left corner of the padding edge
- **border-box** : The background image starts from the upper left corner of the border
- **content-box** : The background image starts from the upper left corner of the content

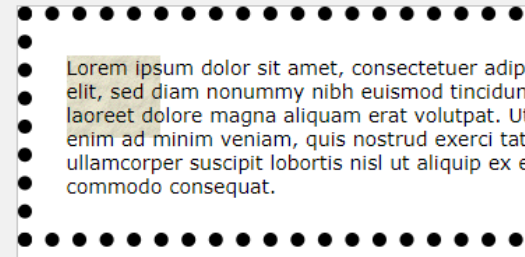
Result:



Result:



Result:



background-clip

- **background-clip** is a new property introduced in CSS3
- The **background-clip** property defines how far the background (color or image) should extend within an element.

Syntax:

background-clip: padding-box|border-box|content-box|text;

It has 3 values:

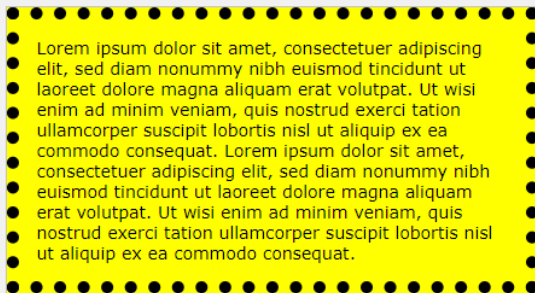
- **padding-box** :
- **border-box**
- **content-box**

background-clip

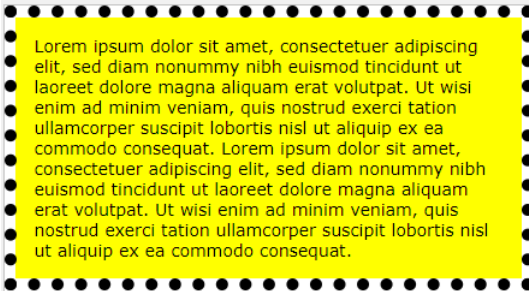
It has 4 values:

- **padding-box** : Default value. The background extends behind the border
- **border-box** : The background extends to the inside edge of the border
- **content-box** : The background extends to the edge of the content box

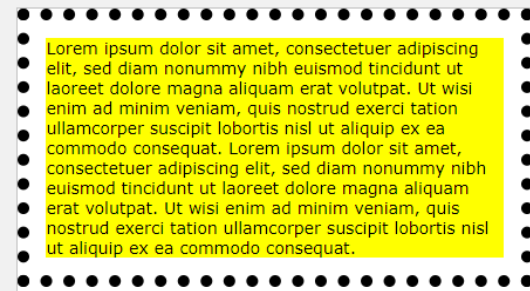
Result:



Result:



Result:

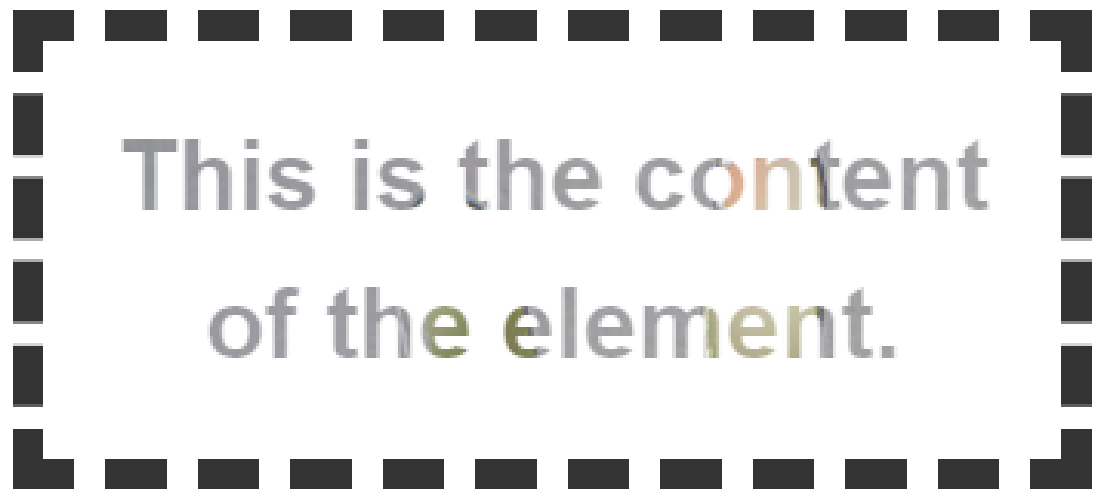


background-clip

It has 4 values:

- **text:** Your background will be clipped in text.

```
{  
    background-clip: text;  
    -webkit-background-clip: text;  
    color: transparent;  
}
```



Advance Text Effect

- CSS Text Effects allows us to apply different types of effect on text used in an HTML document.
- Below are some of the properties in CSS that can be used to add effects to text:
 - text-overflow
 - word-wrap
 - word-break
 - writing-mode

Advance Text Effect : text-overflow

- The CSS **text-overflow** property is a way to limit text that exceeds the width of it's parent.
- It helps to specify the way to represent the portion of overflowing text which is not visible to the user.

It can be clipped:

This is some long text that

or it can be rendered as an ellipsis (...):

This is some long text t...

Advance Text Effect : text-overflow

Syntax:

`text-overflow: clip | ellipsis;`

Value:

clip: Default value for this property. This keyword value will truncate the text at the limit of the content area, therefore the truncation can happen in the middle of a character.

ellipsis: This will display an ellipsis ('...') to represent clipped text. The ellipsis is displayed inside the content area, decreasing the amount of text displayed. If there is not enough space to display the ellipsis, it is clipped.

Advance Text Effect : word-wrap

- The CSS **word-wrap** property defines whether the browser is allowed to line break within words when a word is too long to fit within its parent container.
- If a word is too long to fit within an area, it expands outside:

Syntax:

`word-wrap: break-word;`

Without word-wrap

This paragraph contains a
very long word:
geeksforgeeksforgeeksforgeeksforgeeks

With word-wrap

This paragraph contains a
very long word:
geeksforgeeksforgeeksforg
eeksforgeeks

Advance Text Effect : word-break

- The **word-break** CSS property sets whether line breaks appear wherever the text would otherwise overflow its content box. It specifies line breaking rules.

Syntax:

`word-break: keep-all | break-all;`

Advance Text Effect : word-break

- **break-all** It is used to insert word break between any two characters to prevent word overflow.

word-break: break-all

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- **keep-all:** It is used to break word in default style.

word-break: keep-all

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Advance Text Effect : writing-mode

- The CSS **writing-mode** property specifies whether lines of text are laid out horizontally or vertically.

Syntax

writing-mode: horizontal-tb | vertical-rl;

Other Text Effect

Property	Description
<u>text-align-last</u>	Specifies how to align the last line of a text
<u>text-justify</u>	Specifies how justified text should be aligned and spaced
<u>text-overflow</u>	Specifies how overflowed content that is not displayed should be signaled to the user
<u>word-break</u>	Specifies line breaking rules for non-CJK scripts
<u>word-wrap</u>	Allows long words to be able to be broken and wrap onto the next line
<u>writing-mode</u>	Specifies whether lines of text are laid out horizontally or vertically

2D Transformations

- CSS3 2D transform gives you more freedom to decorate and animate HTML components.
- You have even more features to decorate text and more animation options for decorating div elements.
- CSS3 2D transform contains some basic functions like below:

translate()

rotate()

scale()

skew()

matrix()

2D Transformations

Translate()

Using the `translate()` function, you can move the div and text from their original position to any x-y co-ordinate without affecting other HTML components.

Syntax:

```
transform: translate(x-value,y-value);
```

For example:

```
transform: translate(20px,20px);
```


2D Transformations

Rotate()

Using the `translate()` function, you can move the div and text from their original position to any x-y co-ordinate without affecting other HTML components.

Syntax:

```
transform: rotate(deg);
```

For example:

```
transform: rotate(30deg);
```

2D Transformations

Scale()

It scales the old div to the new size as specified. This scaling takes place in both x and y direction. You can control the amount of scaling by specifying values, as parameters, to scale() function.

Syntax:

```
transform: scale(x-value,y-value);
```

For example:

```
transform: scale(2,3);
```

```
transform: scale(0.5,0.5);
```

2D Transformations

Scale Specific Coordinates

Syntax:

`transform: scaleX(x-value);`

For example:

`transform: scale(2);`

Syntax:

`transform: scaleY(y-value);`

For example:

`transform: scale(3);`

2D Transformations

Skew()

When a skew transform function is applied to an element, the element bends to a specified amount as provided in the parameters list. You have to specify the angle value for x-direction as well as for the y-direction.

Syntax:

```
transform: skew(x-deg,y-deg);
```

For example:

```
transform: skew(10deg,20deg);
```

2D Transformations

SkewSpecific Coordinates

Syntax:

`transform: skewX(x-deg);`

For example:

`transform: skewX(30deg);`

Syntax:

`transform: skewY(y-deg);`

For example:

`transform: skewY (20deg);`

2D Transformations : transform-origin

- This property allows you to set coordinates from where you want to transform object.
- By default transform-origin is set to centre point of element.

■

Syntax:

`transform-origin: x-value, y-value;`

For example:

`transform-origin: right top;`

`transform-origin: 50% 30%;`

2D Transformations

matrix()

- The work of matrix transform function is to combine all the above 2D transform function into one property.
- The values are in the form of linear transformation matrix.

Syntax:

transform: matrix(a, b, c, d, tx, ty)

matrix(scaleX(), skewY(), skewX(), scaleY(), translateX(), translateY())

Making Animation with 2D Transformations

```
#mydiv{
  width: 300px;
  height: 100px;
  top:100px;
  position:absolute;
  background-color: blueviolet;
  transform: rotate(0deg) scale(1);
  transform-origin: left top;
  transition: transform 1s;
}

#mydiv:hover{
  transform: rotate(30deg) scale(2);
  transform-origin: left top;
}
```


3D Transformations

- With CSS3 3D transform feature you can perform basic transform manipulations such as move, rotate, scale and skew on elements in a three-dimensional space.
- Below are the functions that we use for 3d Transform
 - `translate3d()`
 - `rotate3d()`
 - `scale3d()`
 - `skew3d()`
 - `matrix3d()`

3D Transformations

Translate3d()

Using the `translate()` function, you can move the div and text from their original position to any x-y co-ordinate without affecting other HTML components.

Syntax:

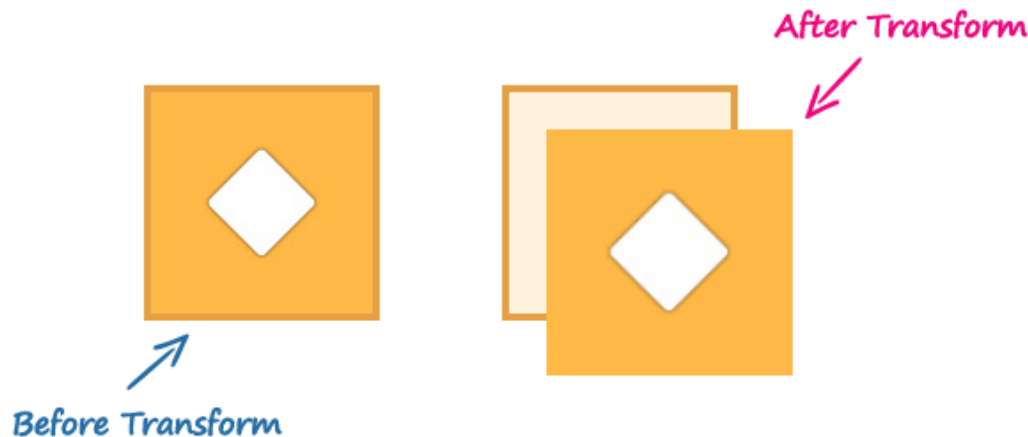
```
transform: translate3d(tx,ty,tz);
```

For example:

```
transform: translate(20px,20px,20px);
```

3D Transformations

- The 3D transform however uses the three-dimensional coordinate system, but movement along the Z-direction is not always noticeable because the elements exist on a two-dimensional plane (a flat surface) and have no depth.
- The perspective and perspective-origin CSS properties can be used to add a feeling of depth to a scene by making the elements higher on the Z-axis



3D Transformations

Rotate3d()

The rotate3d() function rotates the element in 3D space by the specified angle around the [x,y,z] direction vector. This can be written as rotate(vx, vy, vz, angle).

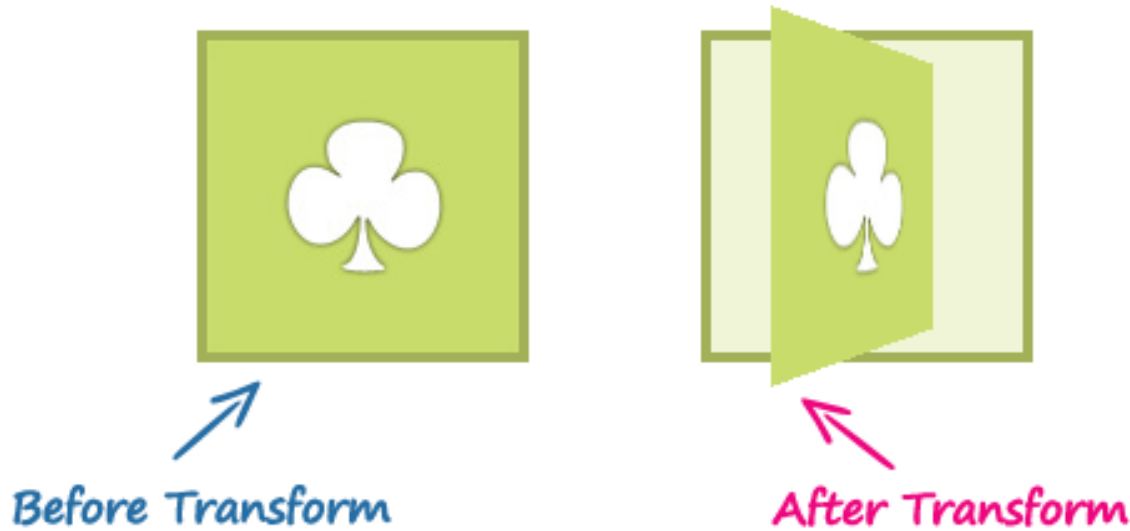
Syntax:

```
transform: rotate3d(vx,vy,vz,deg);
```

3D Transformations

For example:

```
transform: rotate3d(0,1,0,60deg);
```



The function `rotate3d(0, 1, 0, 60deg)` rotates the image along the Y-axis by the angle 60 degrees. You can use negative values to rotate the element in opposite direction

3D Transformations

Scale3d()

The `scale3d()` function changes the size of an element. It can be written as `scale(sx, sy, sz)`.

Syntax:

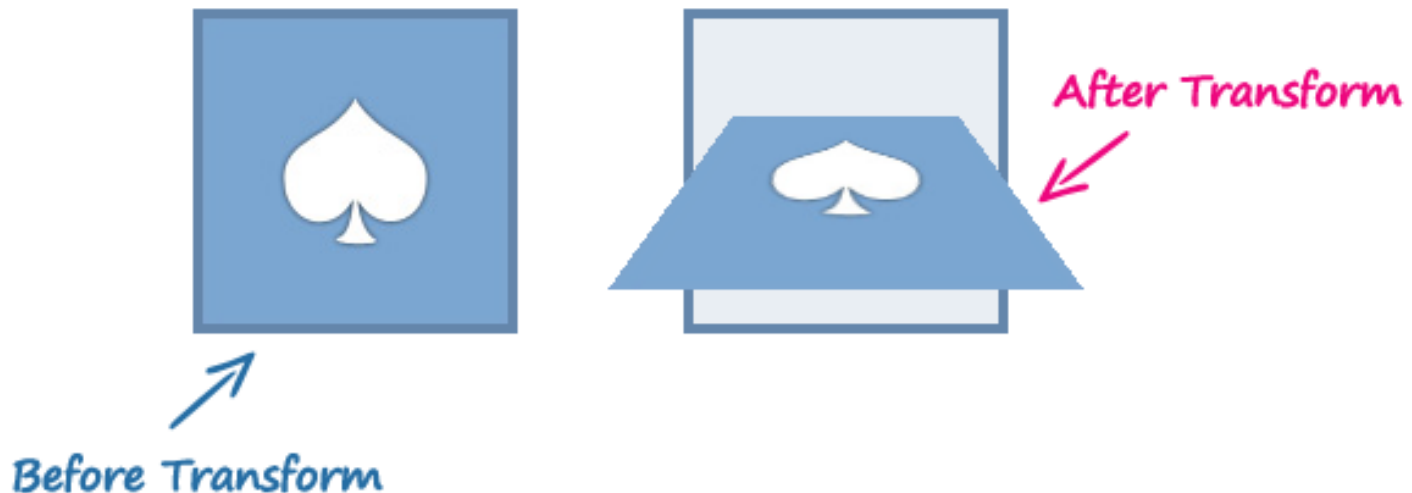
```
transform: scale3d(sx,sy,sz);
```

For example:

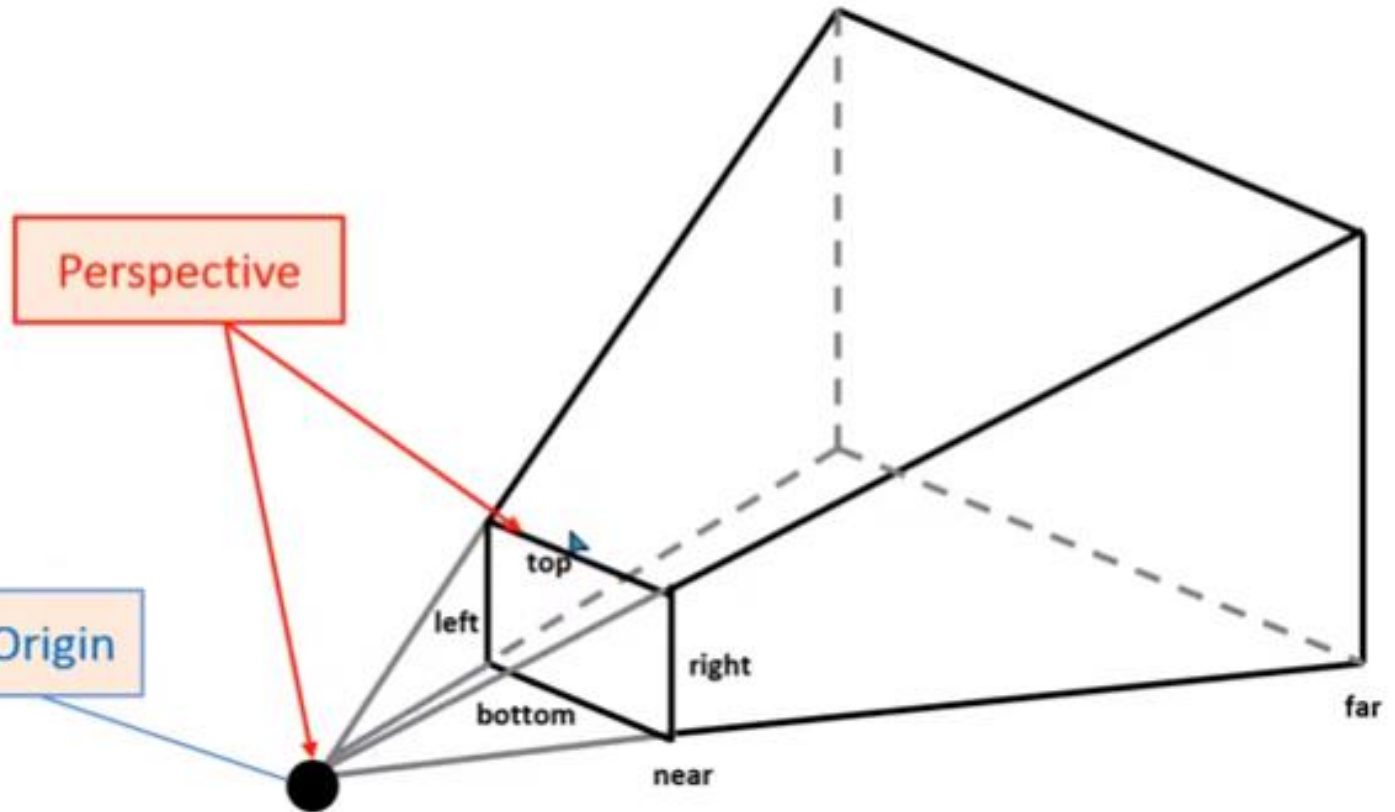
```
transform: scale3d(1, 1, 2);
```

3D Transformations

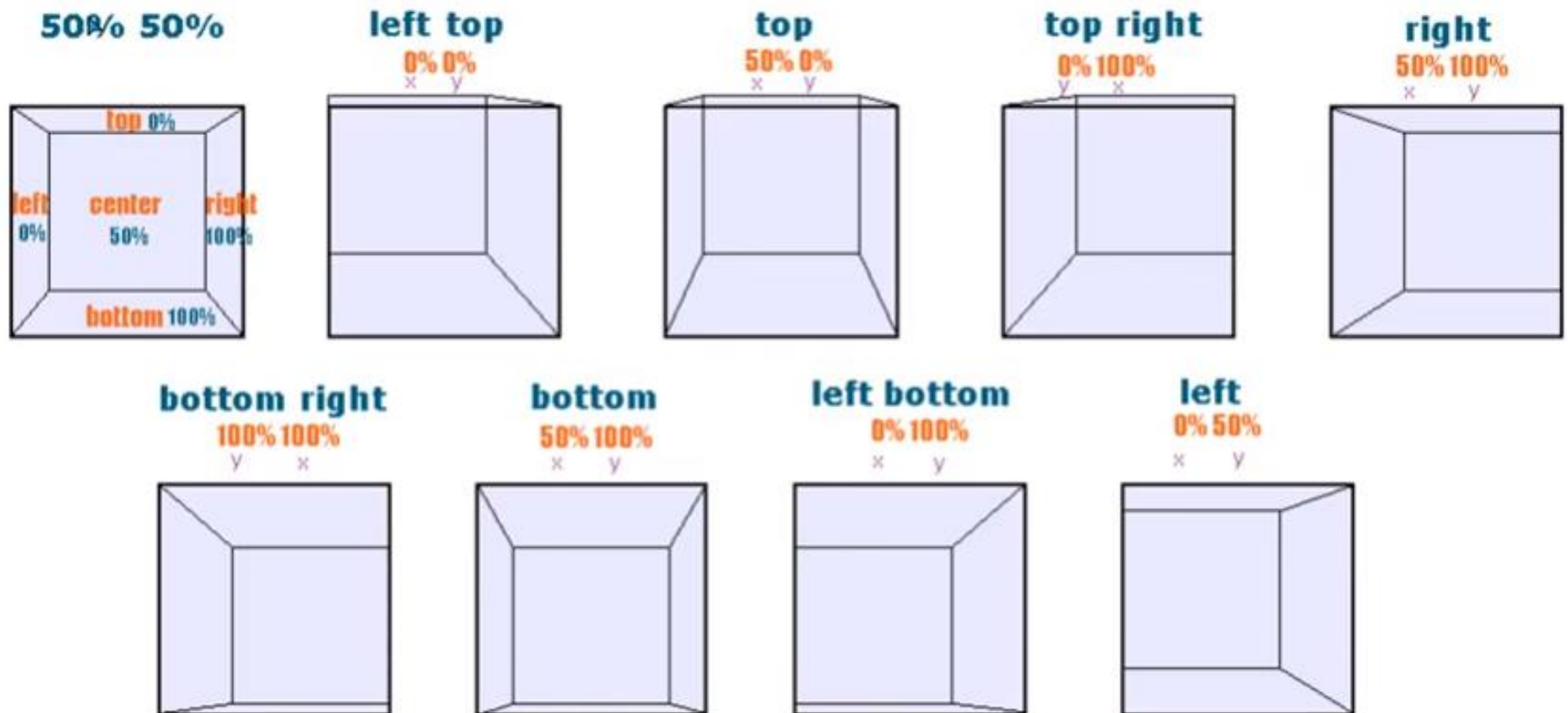
```
transform: scale3d(1, 1, 2) rotate3d(1, 0, 0, 60deg);
```



Perspective-Origin



Perspective Property



Adding Transitions to Elements

The transition property in CSS is used to make some transition effect. The **transition** is the combination of four properties which are listed below:

- transition-property
- transition-duration
- transition-timing-function
- transition-delay

Note: The transition effect can be defined in two states (hover and active) using pseudo-classes like **:hover** or **:active** or classes dynamically set by using JavaScript.

Adding Transitions to Elements

Property Values:

- **transition-property:** It specifies the CSS properties to which a transition effect should be applied.
- **transition-duration:** It specifies the length of time a transition animation should take to complete.
- **transition-timing-function:** It specifies the speed of transition.
- **transition-delay:** It specifies the transition delay or time when transition starts.

Adding Transitions to Elements

Syntax:

*transition: transition-property transition-duration
transition-timing-function transition-delay;*

Example:

```
<style>
  div {
    width: 100px;
    height: 270px;
    background: green;
    transition: width 5s ease-in .2s;
    display: inline-block;
  }

  div:hover {
    width: 300px;
  }
</style>
```

CSS Animation

CSS3 allows animation of HTML elements without using JavaScript or Flash!

An animation lets an element gradually change from one style to another.

You can change as many CSS properties you want, as many times you want.

To use CSS animation, you must first specify some keyframes for the animation.

Keyframes hold what styles the element will have at certain times.

CSS Animation

Properties of Animation

- **@keyframes:** Keyframes hold what styles the element will have at certain times.
- **animation-name:** Specifies the name of @keyframes defined animations that should be applied to the selected element.
- **animation-duration:** Specifies how many seconds or milliseconds that an animation takes to complete one cycle of the animation.
- **animation-delay:** Specifies a delay before the animation will start.

CSS Animation

- **animation-iteration-count:** Specifies the number of times an animation cycle should be played before stopping.
- **animation-direction:** Specifies whether or not the animation should play in reverse on alternate cycles.
- **animation-timing-function:** Specifies how the animation will progress over the duration of each cycle i.e. the easing functions.
- **animation-fill-mode:** Specifies how a CSS animation should apply styles to its target before and after it is executing.
- **Animation:** Used for shorthand.

CSS Animation Shorthand

- **animation:** Property used as shorthand for all above property value.

Syntax:

*animation: name duration timing-function delay
iteration-count direction fill-mode*

CSS3 flexbox

- Before the Flexbox Layout module, there were four layout modes:
 - **Block**, for sections in a webpage
 - **Inline**, for text
 - **Table**, for two-dimensional table data
 - **Positioned**, for explicit position of an element

CSS3 flexbox

- Flexbox means Flexible Box, it is introduced in CSS3 only.
- Flexbox is a layout module, make it easier to design flexible responsive layout structure without using float and positioning.

How to use flexbox?

- To start using the Flexbox model, you need to first define a flex container.

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```
.flex-container {  
  display: flex;  
  background-color:  DodgerBlue;  
}  
  
.flex-container>div {  
  background-color:  #f1f1f1;  
  margin: 10px;  
  padding: 20px;  
  font-size: 30px;  
}
```

```
<div class="flex-container">  
  <div>1</div>  
  <div>2</div>  
  <div>3</div>  
</div>
```

How to use flexbox?

- A Flexible Layout must have a parent element with the *display* property set to *flex*.
- Direct child elements(s) of the flexible container automatically becomes flexible items.
- Flexbox property are as below:
 - flex-direction
 - The flex-direction property defines in which direction the container wants to stack the flex items.

How to use flexbox?

- flex-wrap
 - The flex-wrap property specifies whether the flex items should wrap or not.
- flex-flow
 - The flex-flow property is a shorthand property for setting both the flex-direction and flex-wrap properties.
- justify-content
 - The justify-content property is used to align the flex items:
- align-items
 - The align-items property is used to align the flex items vertically.
- align-content
 - The align-content property is used to align the flex lines.

Flex Item Property

- Flex Item Properties are as below
 - order
 - flex-grow
 - flex-shrink
 - flex-basis
 - Flex:flex-grow shrink basis

CSS Filters

- `blur()`
- `brightness()`
- `contrast()`
- `drop-shadow()`
- `grayscale()`
- `hue-rotate()`
- `invert()`
- `opacity()`
- `saturate()`
- `sepia()`